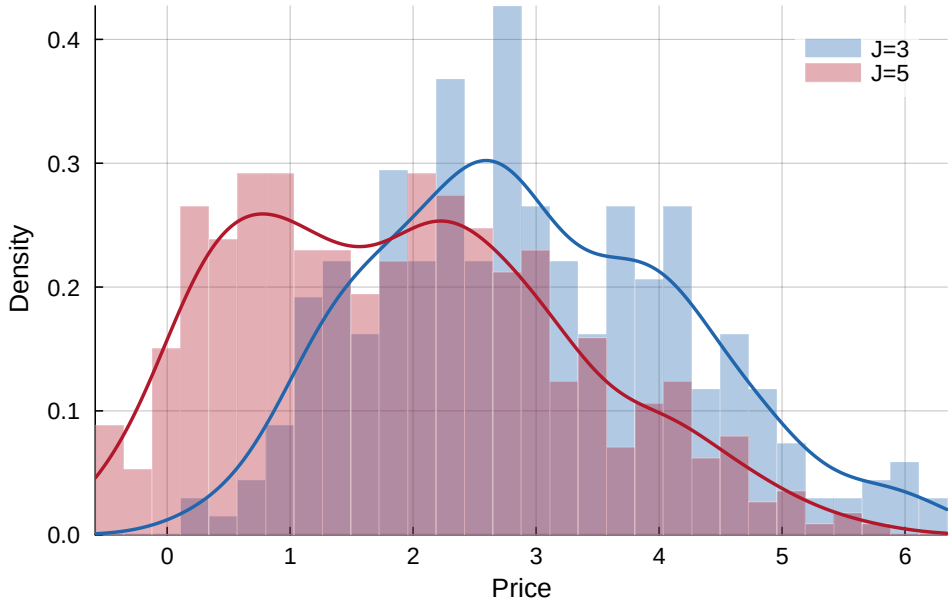
# Economics 450-1, Problem Set 1: Figures and Tables

## Question 6: Market simulation and welfare

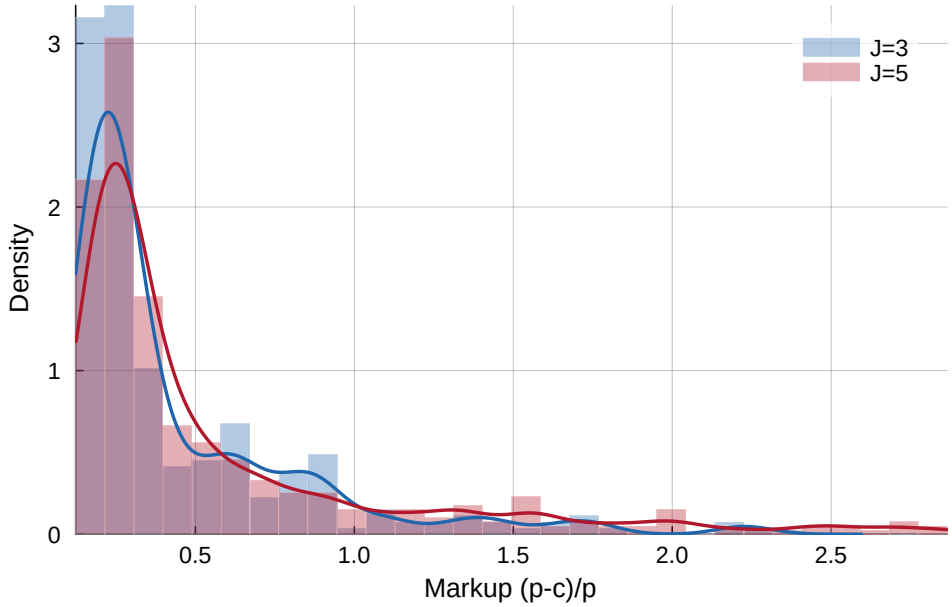
### 6a

Figure 1: Distribution of price, $J = 3$ versus $J = 5$ (tails beyond the 1st and 99th percentiles not drawn).



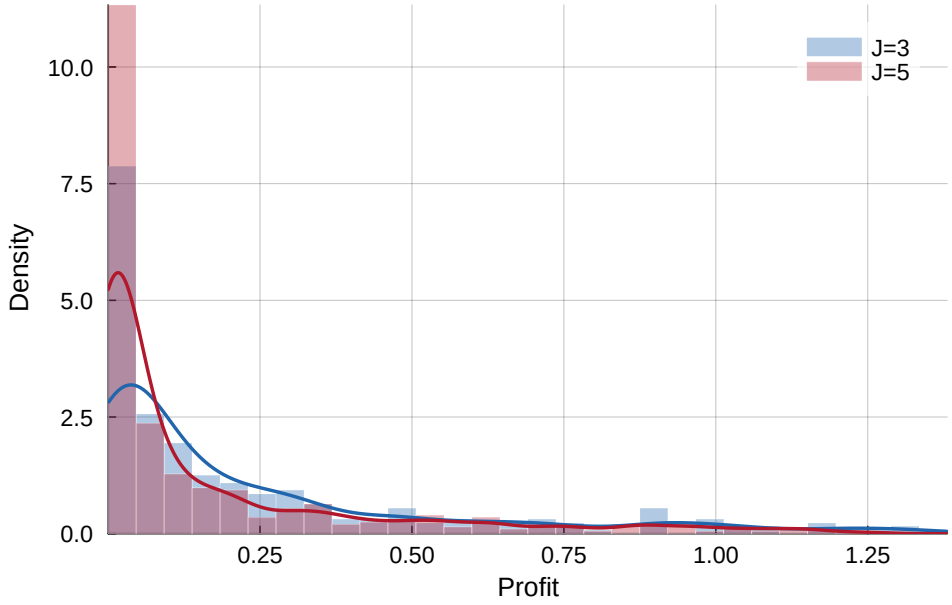

Source: true-parameter simulation of `100markets3products.mat` and `100markets5products.mat` with common consumer draws.

Figure 2: Distribution of markup, $J = 3$ versus $J = 5$ (tails beyond the 5th and 95th percentiles not drawn).




Source: true-parameter simulation of `100markets3products.mat` and `100markets5products.mat` with common consumer draws.

Figure 3: Distribution of profit, $J = 3$ versus $J = 5$ (tails beyond the 5th and 95th percentiles not drawn).




Source: true-parameter simulation of `100markets3products.mat` and `100markets5products.mat` with common consumer draws.

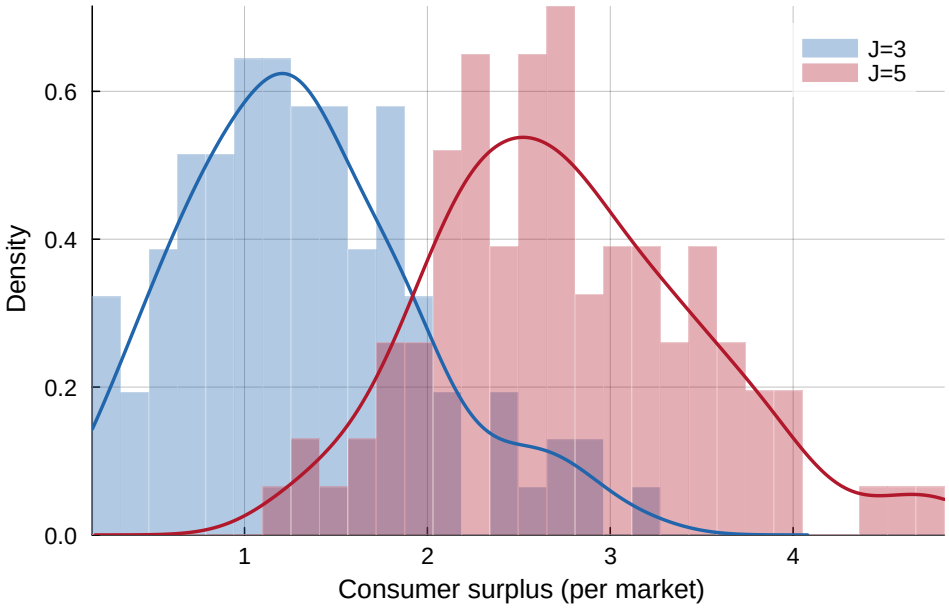
Table 1: Descriptive statistics of prices, markups and profits (all products and markets).

| Variable | $J$ | Mean | SD | Median | Q25 | Q75 |
|---|---|---|---|---|---|---|
| Price | 3 | 3.027 | 1.371 | 2.828 | 1.988 | 3.964 |
| Price | 5 | 1.891 | 1.435 | 1.816 | 0.780 | 2.830 |
| Markup | 3 | 0.472 | 0.634 | 0.274 | 0.197 | 0.574 |
| Markup | 5 | −0.856 | 38.116 | 0.315 | 0.222 | 0.782 |
| Profit | 3 | 0.309 | 0.463 | 0.105 | 0.021 | 0.353 |
| Profit | 5 | 0.254 | 0.514 | 0.041 | 0.004 | 0.225 |

Source: true-parameter simulation of `100markets3products.mat` and `100markets5products.mat` with common consumer draws; full sample, no trimming.

## 6b

Figure 4: Distribution of consumer surplus across markets, $J = 3$ versus $J = 5$.




Source: true-parameter simulation of `100markets3products.mat` and `100markets5products.mat` with common consumer draws; expected surplus averaged over the simulated consumers.

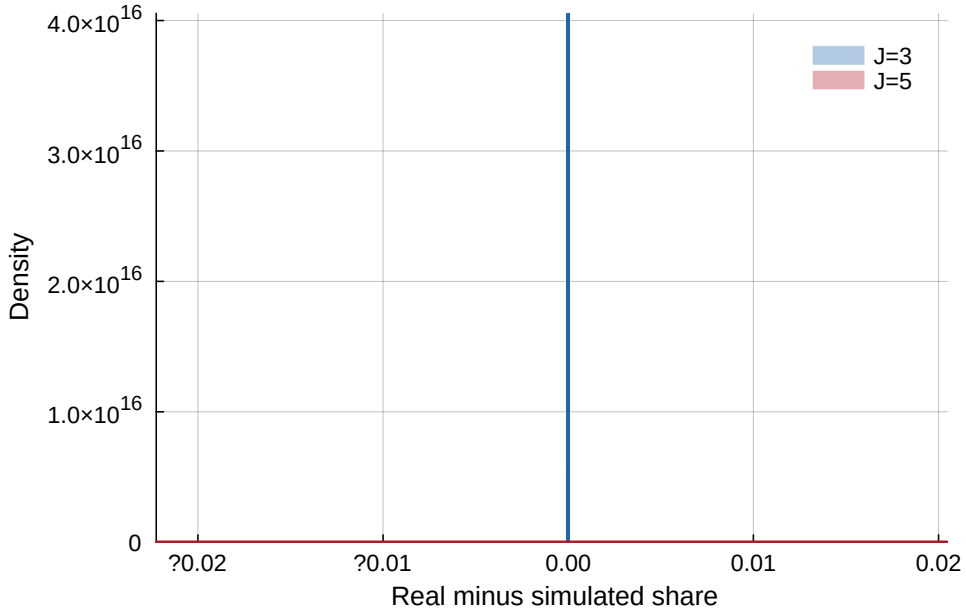
Table 2: Descriptive statistics of consumer surplus across markets.

| $J$ | Mean | SD | Median | Q25 | Q75 |
|---|---|---|---|---|---|
| 3 | 1.314 | 0.642 | 1.251 | 0.851 | 1.729 |
| 5 | 2.765 | 0.747 | 2.697 | 2.258 | 3.228 |

Source: true-parameter simulation of `100markets3products.mat` and `100markets5products.mat` with common consumer draws.

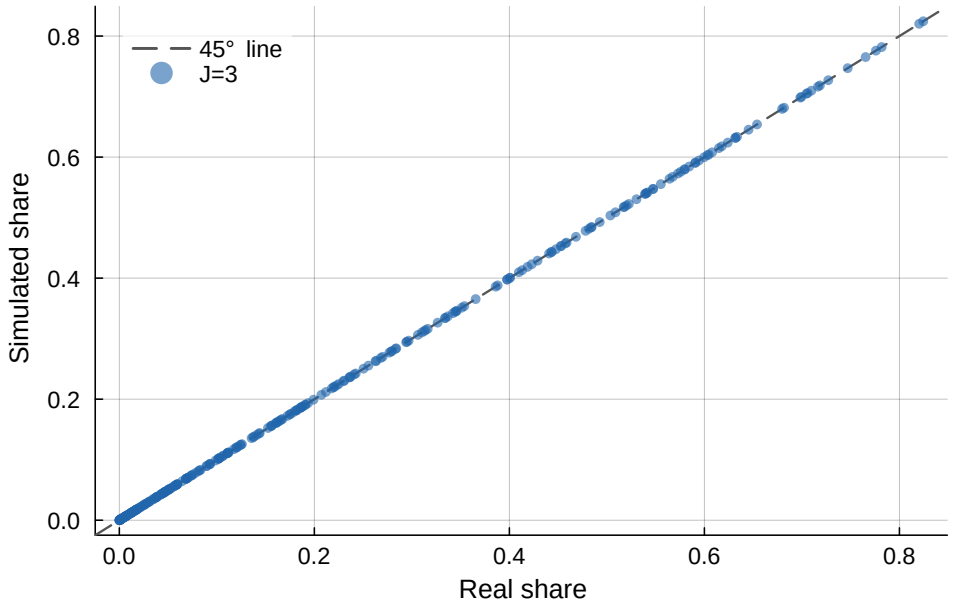
**6c**

Figure 5: Difference between real and simulated shares, $J = 3$ versus $J = 5$.



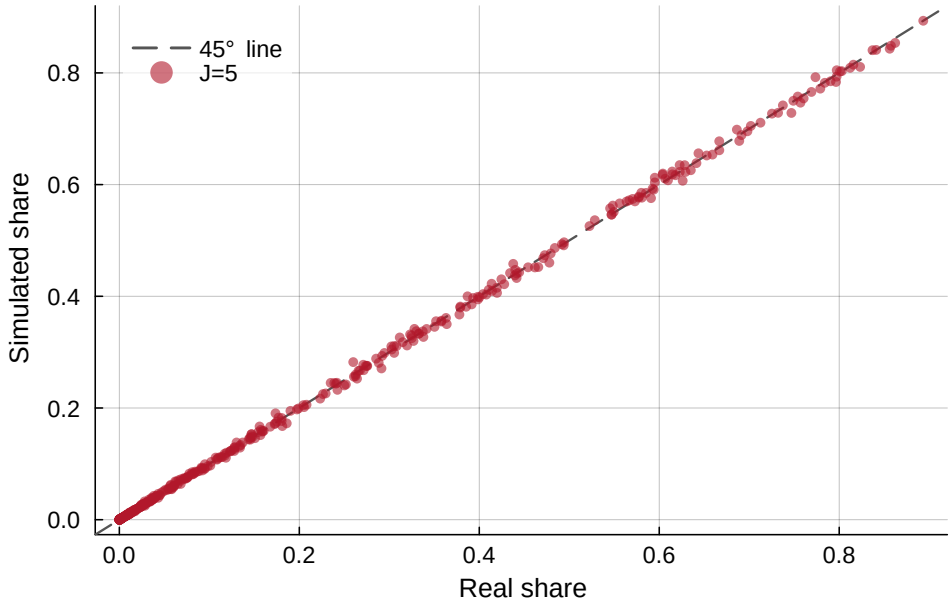

Source: true-parameter simulation of `100markets3products.mat` and `100markets5products.mat` with common consumer draws.

Figure 6: Real versus simulated shares, $J = 3$.




Source: true-parameter simulation of `100markets3products.mat` and `100markets5products.mat` with common consumer draws.

Figure 7: Real versus simulated shares, $J = 5$.




Source: true-parameter simulation of `100markets3products.mat` and `100markets5products.mat` with common consumer draws.

Table 3: Discrepancy between real and simulated shares (real minus simulated), and maximum deviation of simulated shares plus outside share from one.

| $J$ | Mean diff. | Mean abs. diff. | RMSE | Correlation | Max $\lvert \sum_j s_{jt} + s_{0t} - 1 \rvert$ |
|---|---|---|---|---|---|
| 3 | −0.00000 | 0.00000 | 0.00000 | 1.00000 | $9.99 \times 10^{-16}$ |
| 5 | 0.00001 | 0.00264 | 0.00481 | 0.99980 | $1.33 \times 10^{-15}$ |

Source: true-parameter simulation of `100markets3products.mat` and `100markets5products.mat` with common consumer draws.

## Question 7: Demand moments and instruments

### 7a

Table 4: Sample analogues of the moments $E[\xi_{jt} z_{jt}]$ and $t$-tests of $H_0 : E[\xi_{jt} z_{jt}] = 0$.

| Moment | Mean | SD | $t$-stat | $p$-value | $N$ |
|---|---|---|---|---|---|
| $E[\xi_{jt} \cdot 1]$ | 0.0435 | 1.0037 | 0.750 | 0.454 | 300 |
| $E[\xi_{jt} x_{2jt}]$ | 0.0207 | 0.5822 | 0.617 | 0.538 | 300 |
| $E[\xi_{jt} x_{3jt}]$ | 0.0355 | 0.9649 | 0.638 | 0.524 | 300 |
| $E[\xi_{jt} p_{jt}]$ | 0.2950 | 3.6702 | 1.392 | 0.165 | 300 |
| $E[\xi_{jt} \bar{p}_{j,-t}]$ | 0.1437 | 3.1692 | 0.785 | 0.433 | 300 |

Source: true structural errors $\xi_{jt}$ and data of `100markets3products.mat`.

### 7c

Table 5: Sample analogues of the moments with the BLP rival-characteristic sums and $t$-tests of $H_0 : E[\xi_{jt} z_{jt}] = 0$.

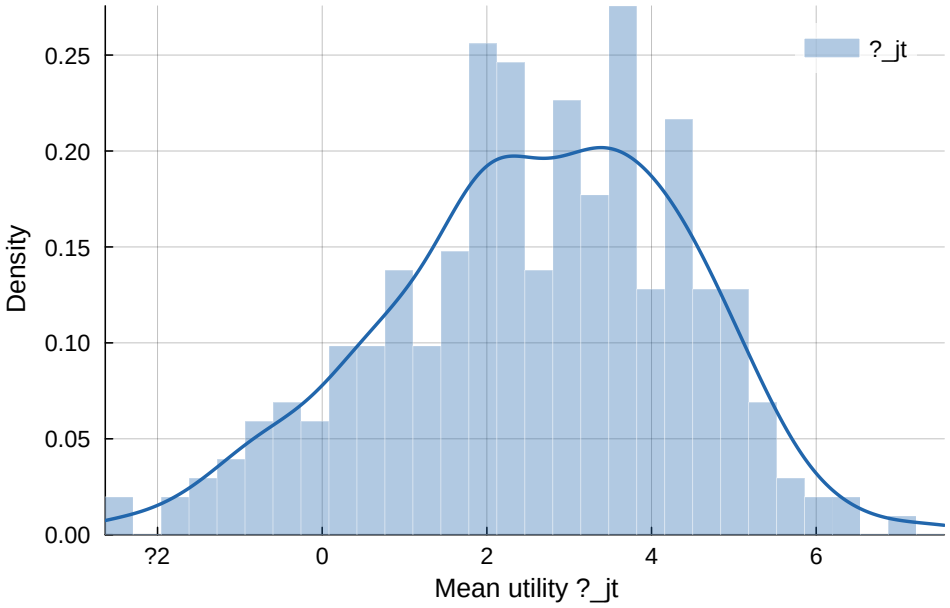
| Moment | Mean | SD | $t$-stat | $p$-value | $N$ |
|---|---|---|---|---|---|
| $E[\xi_{jt} \sum_{k \neq j} x_{2kt}]$ | 0.0457 | 1.1542 | 0.686 | 0.493 | 300 |
| $E[\xi_{jt} \sum_{k \neq j} x_{3kt}]$ | −0.0087 | 1.2301 | −0.123 | 0.902 | 300 |

Source: true structural errors $\xi_{jt}$ and data of `100markets3products.mat`.

## Question 8: Mixed logit by nested fixed point

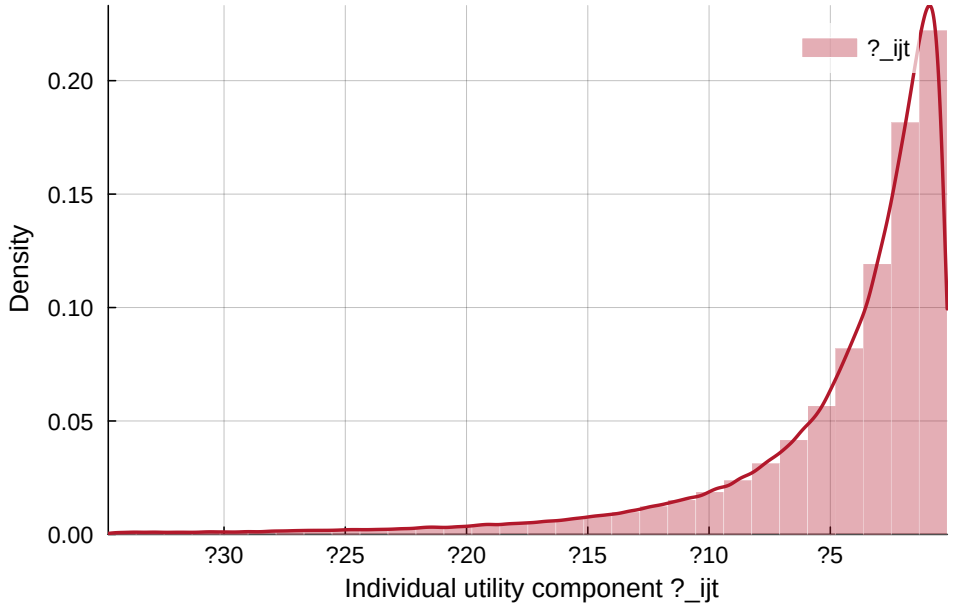
### 8a

Figure 8: Distribution of mean utilities $\delta_{jt}$ at the true parameters.




Source: true-parameter simulation of `100markets3products.mat`; $\delta_{jt} = x'_{jt}\beta - \bar{\alpha} p_{jt} + \xi_{jt}$, $\mu_{ijt} = -\sigma_\alpha \nu_i p_{jt}$.

Figure 9: Distribution of $\mu_{ijt}$ over products, markets and simulated consumers at the true parameters (tails beyond the 1st and 99th percentiles not drawn).




Source: true-parameter simulation of `100markets3products.mat`; $\delta_{jt} = x'_{jt}\beta - \bar{\alpha}p_{jt} + \xi_{jt}$, $\mu_{ijt} = -\sigma_\alpha \nu_i p_{jt}$.

## 8c

Table 6: Estimates by starting point, T=100, J=3, BLP instruments (standard errors in parentheses; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$).

| Start $\sigma_0$ | Constant | $x_2$ | $x_3$ | Price | $\sigma_\alpha$ | GMM obj. |
|---|---|---|---|---|---|---|
| 0.594 | 5.589 (n.a.) | 1.100 (n.a.) | 1.082 (n.a.) | −1.168 (n.a.) | 1.054 (n.a.) | $5.23 \times 10^{-3}$ |
| 0.908 | 5.589 ($1.44 \times 10^7$) | 1.100 ($3.17 \times 10^6$) | 1.082 ($1.06 \times 10^6$) | −1.168 ($4.67 \times 10^6$) | 1.054 ($1.10 \times 10^6$) | $5.23 \times 10^{-3}$ |
| 0.859 | 5.589 ($2.59 \times 10^6$) | 1.100 ($5.71 \times 10^5$) | 1.082 ($1.91 \times 10^5$) | −1.168 ($8.41 \times 10^5$) | 1.054 ($1.98 \times 10^5$) | $5.23 \times 10^{-3}$ |
| 1.426 | 5.589 ($4.30 \times 10^6$) | 1.100 ($9.47 \times 10^5$) | 1.082 ($3.18 \times 10^5$) | −1.168 ($1.40 \times 10^6$) | 1.054 ($3.29 \times 10^5$) | $5.23 \times 10^{-3}$ |
| 0.182 | 5.589 (n.a.) | 1.100 (n.a.) | 1.082 (n.a.) | −1.168 (n.a.) | 1.054 (n.a.) | $5.23 \times 10^{-3}$ |
| 0.876 | 5.589 ($1.10 \times 10^7$) | 1.100 ($2.42 \times 10^6$) | 1.082 ($8.12 \times 10^5$) | −1.168 ($3.56 \times 10^6$) | 1.054 ($8.39 \times 10^5$) | $5.23 \times 10^{-3}$ |
| 0.254 | 5.589 (n.a.) | 1.100 (n.a.) | 1.082 (n.a.) | −1.168 (n.a.) | 1.054 (n.a.) | $5.23 \times 10^{-3}$ |
| 1.064 | 5.589 ($9.62 \times 10^7$) | 1.100 ($2.12 \times 10^7$) | 1.082 ($7.12 \times 10^6$) | −1.168 ($3.13 \times 10^7$) | 1.054 ($7.36 \times 10^6$) | $5.23 \times 10^{-3}$ |
| **0.228** | **5.589** ($1.62 \times 10^7$) | **1.100** ($3.58 \times 10^6$) | **1.082** ($1.20 \times 10^6$) | **−1.168** ($5.27 \times 10^6$) | **1.054** ($1.24 \times 10^6$) | $5.23 \times 10^{-3}$ |
| 1.168 | 5.589 (n.a.) | 1.100 (n.a.) | 1.082 (n.a.) | −1.168 (n.a.) | 1.054 (n.a.) | $5.23 \times 10^{-3}$ |
| 0.768 | 5.589 (n.a.) | 1.100 (n.a.) | 1.082 (n.a.) | −1.168 (n.a.) | 1.054 (n.a.) | $5.23 \times 10^{-3}$ |
| 1.378 | 5.589 ($2.01 \times 10^7$) | 1.100 ($4.43 \times 10^6$) | 1.082 ($1.49 \times 10^6$) | −1.168 ($6.53 \times 10^6$) | 1.054 ($1.54 \times 10^6$) | $5.23 \times 10^{-3}$ |
| 1.088 | 5.589 ($2.59 \times 10^7$) | 1.100 ($5.71 \times 10^6$) | 1.082 ($1.92 \times 10^6$) | −1.168 ($8.41 \times 10^6$) | 1.054 ($1.98 \times 10^6$) | $5.23 \times 10^{-3}$ |
| 0.770 | 5.589 (n.a.) | 1.100 (n.a.) | 1.082 (n.a.) | −1.168 (n.a.) | 1.054 (n.a.) | $5.23 \times 10^{-3}$ |
| 0.607 | 5.589 ($2.25 \times 10^7$) | 1.100 ($4.97 \times 10^6$) | 1.082 ($1.67 \times 10^6$) | −1.168 ($7.32 \times 10^6$) | 1.054 ($1.72 \times 10^6$) | $5.23 \times 10^{-3}$ |
| 0.444 | 5.589 (n.a.) | 1.100 (n.a.) | 1.082 (n.a.) | −1.168 (n.a.) | 1.054 (n.a.) | $5.23 \times 10^{-3}$ |
| True | 5.000 | 1.000 | 1.000 | −1.000 | 1.000 | |

Source: author's calculations from the stored full estimates `q8c_blp_only_T100_J3` (per-start values recomputed at each stored $\sigma$ minimizer, no re-optimization). Bold: start with the lowest GMM objective across starts. (n.a.): estimated variance not positive; very large standard errors indicate weak identification..

Table 7: Convergence criterion and contraction error by starting point, T=100, J=3.

| Start $\sigma_0$ | $\hat{\sigma}_\alpha$ | GMM obj. | Converged | Iterations | $\|\nabla\|_\infty$ | Contraction error |
|---|---|---|---|---|---|---|
| 0.594 | 1.0543 | $5.23 \times 10^{-3}$ | yes | 3 | $1.30 \times 10^{-10}$ | $5.74 \times 10^{-11}$ |
| 0.908 | 1.0543 | $5.23 \times 10^{-3}$ | yes | 3 | $3.33 \times 10^{-11}$ | $5.74 \times 10^{-11}$ |
| 0.859 | 1.0543 | $5.23 \times 10^{-3}$ | yes | 3 | $4.75 \times 10^{-11}$ | $5.74 \times 10^{-11}$ |
| 1.426 | 1.0543 | $5.23 \times 10^{-3}$ | yes | 3 | $5.75 \times 10^{-10}$ | $5.74 \times 10^{-11}$ |
| 0.182 | 1.0543 | $5.23 \times 10^{-3}$ | yes | 3 | $2.33 \times 10^{-11}$ | $5.74 \times 10^{-11}$ |
| 0.876 | 1.0543 | $5.23 \times 10^{-3}$ | yes | 3 | $4.18 \times 10^{-11}$ | $5.74 \times 10^{-11}$ |
| 0.254 | 1.0543 | $5.23 \times 10^{-3}$ | yes | 3 | $4.98 \times 10^{-11}$ | $5.74 \times 10^{-11}$ |
| 1.064 | 1.0543 | $5.23 \times 10^{-3}$ | yes | 2 | $2.23 \times 10^{-9}$ | $5.74 \times 10^{-11}$ |
| **0.228** | **1.0543** | $5.23 \times 10^{-3}$ | **yes** | **3** | $3.90 \times 10^{-11}$ | $5.74 \times 10^{-11}$ |
| 1.168 | 1.0543 | $5.23 \times 10^{-3}$ | yes | 3 | $3.75 \times 10^{-11}$ | $5.74 \times 10^{-11}$ |
| 0.768 | 1.0543 | $5.23 \times 10^{-3}$ | yes | 3 | $7.73 \times 10^{-11}$ | $5.74 \times 10^{-11}$ |
| 1.378 | 1.0543 | $5.23 \times 10^{-3}$ | yes | 3 | $3.59 \times 10^{-10}$ | $5.74 \times 10^{-11}$ |
| 1.088 | 1.0543 | $5.23 \times 10^{-3}$ | yes | 2 | $8.28 \times 10^{-9}$ | $5.74 \times 10^{-11}$ |
| 0.770 | 1.0543 | $5.23 \times 10^{-3}$ | yes | 3 | $7.64 \times 10^{-11}$ | $5.74 \times 10^{-11}$ |
| 0.607 | 1.0543 | $5.23 \times 10^{-3}$ | yes | 3 | $1.28 \times 10^{-10}$ | $5.74 \times 10^{-11}$ |
| 0.444 | 1.0543 | $5.23 \times 10^{-3}$ | yes | 3 | $1.22 \times 10^{-10}$ | $5.74 \times 10^{-11}$ |

Source: author's calculations from the stored full estimates `q8c_blp_only_T100_J3` (per-start values recomputed at each stored $\sigma$ minimizer, no re-optimization); failed starts from `q8c_blp_only_T100_J3_failed_starts`. Bold: start with the lowest GMM objective across starts..

Table 8: Estimates with and without the Hausman instrument, T=100, J=3.

| | True | BLP | BLP + Hausman |
|---|---|---|---|
| Constant | 5.000 | 5.589 | $4.983^{***}$ |
| | | $(1.62 \times 10^7)$ | (0.404) |
| $x_2$ (uniform) | 1.000 | 1.100 | $0.967^{***}$ |
| | | $(3.58 \times 10^6)$ | (0.225) |
| $x_3$ (normal) | 1.000 | 1.082 | $1.037^{***}$ |
| | | $(1.20 \times 10^6)$ | (0.076) |
| Price coefficient | −1.000 | −1.168 | $-0.972^{***}$ |
| | | $(5.27 \times 10^6)$ | (0.116) |
| $\sigma_\alpha$ | 1.000 | 1.054 | $1.007^{***}$ |
| | | $(1.24 \times 10^6)$ | (0.080) |
| GMM objective | | $5.23 \times 10^{-3}$ | $5.56 \times 10^{-3}$ |
| Observations ($JT$) | | 300 | 300 |

Source: author's calculations from the stored full estimates `q8c_blp_only_T100_J3` and `q8c_blp_hausman_T100_J3`; best start of each. (n.a.): estimated variance not positive; very large standard errors indicate weak identification..

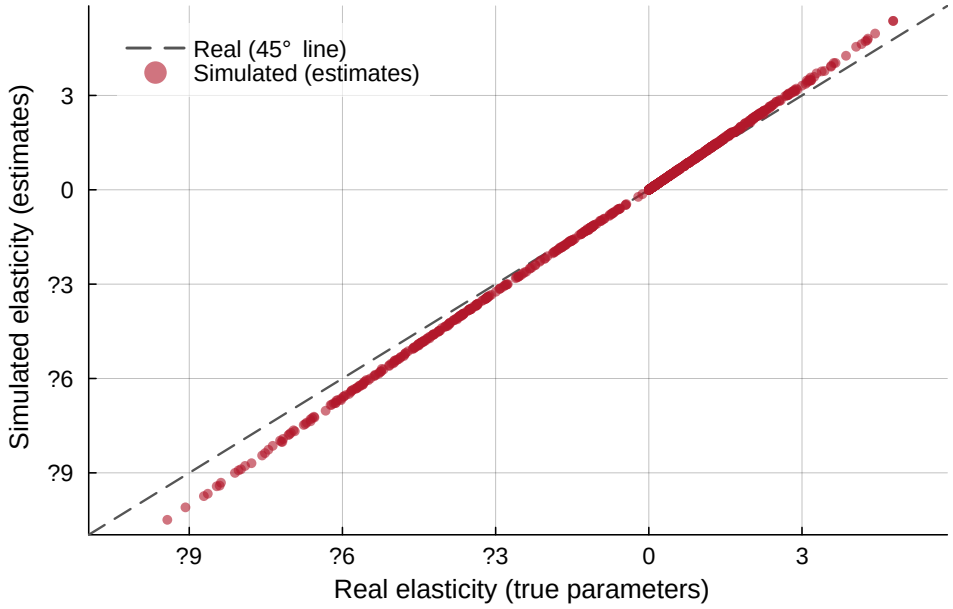
Table 9: First stage of price on the instruments, T=100, J=3 (heteroskedasticity-robust standard errors in parentheses; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$; $F$ tests the excluded instruments given $x$).

| Dependent variable: price | BLP | BLP + Hausman |
|---|---|---|
| Constant | 2.928*** | 0.145 |
| | (0.251) | (0.423) |
| $x_2$ | 0.550** | 0.319 |
| | (0.269) | (0.255) |
| $x_3$ | 0.111 | 0.057 |
| | (0.074) | (0.072) |
| BLP rival sum $x_2$ | −0.174 | −0.057 |
| | (0.182) | (0.171) |
| BLP rival sum $x_3$ | −0.131** | −0.104* |
| | (0.061) | (0.054) |
| Hausman $\bar{p}_{j,-t}$ | | 0.919*** |
| | | (0.126) |
| $R^2$ | 0.037 | 0.204 |
| $F$ (excluded instruments) | 2.9 | 22.9 |
| Observations | 300 | 300 |

Source: author's calculations from `100markets3products.mat`; regressions include a constant, $x_2$ and $x_3$.

## 8d

Figure 10: Price elasticities at the estimates versus at the true parameters (all own- and cross-price elements).




Source: author's calculations: elasticities at the estimates of `q8c_blp_only_T100_J3` versus the true-parameter simulation, `100markets3products.mat`.

Table 10: Elasticity of the share of product $j$ with respect to the price of product $k$: real versus simulated (at the estimates), mean and standard deviation across markets.

| | Real | | Simulated | | Simulated − real | |
|---|---|---|---|---|---|---|
| Element $(j,k)$ | Mean | SD | Mean | SD | Mean | SD |
| $(1,1)$ | −5.097 | 1.938 | −5.608 | 2.181 | −0.511 | 0.245 |
| $(1,2)$ | 1.395 | 1.013 | 1.549 | 1.129 | 0.154 | 0.117 |
| $(1,3)$ | 1.000 | 0.884 | 1.111 | 0.984 | 0.111 | 0.100 |
| $(2,1)$ | 0.593 | 0.696 | 0.659 | 0.774 | 0.065 | 0.078 |
| $(2,2)$ | −2.622 | 1.666 | −2.843 | 1.845 | −0.221 | 0.181 |
| $(2,3)$ | 0.985 | 0.917 | 1.087 | 1.014 | 0.101 | 0.098 |
| $(3,1)$ | 0.602 | 0.700 | 0.668 | 0.777 | 0.066 | 0.078 |
| $(3,2)$ | 1.441 | 1.061 | 1.592 | 1.175 | 0.151 | 0.116 |
| $(3,3)$ | −3.407 | 1.819 | −3.714 | 2.021 | −0.307 | 0.204 |

Source: author's calculations: elasticities at the estimates of `q8c_blp_only_T100_J3` versus the true-parameter simulation, `100markets3products.mat`.

### 8e

Table 11: Estimates by starting point, T=10, J=3, BLP instruments (standard errors in parentheses; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$).

| Start $\sigma_0$ | Constant | $x_2$ | $x_3$ | Price | $\sigma_\alpha$ | GMM obj. |
|---|---|---|---|---|---|---|
| 0.594 | 3.198 (3.046) | 0.425 (0.937) | 1.164* (0.653) | −0.634 (1.261) | 0.100 (2.625) | $1.20 \times 10^{-1}$ |
| 0.908 | 3.198 (3.046) | 0.425 (0.937) | 1.164* (0.653) | −0.634 (1.261) | 0.100 (2.625) | $1.20 \times 10^{-1}$ |
| 0.859 | 3.198 (3.046) | 0.425 (0.937) | 1.164* (0.653) | −0.634 (1.261) | 0.100 (2.625) | $1.20 \times 10^{-1}$ |
| 1.426 | 3.198 (3.046) | 0.425 (0.937) | 1.164* (0.653) | −0.634 (1.261) | 0.100 (2.625) | $1.20 \times 10^{-1}$ |
| 0.182 | 3.198 (3.046) | 0.425 (0.937) | 1.164* (0.653) | −0.634 (1.261) | 0.100 (2.625) | $1.20 \times 10^{-1}$ |
| 0.876 | 3.198 (3.046) | 0.425 (0.937) | 1.164* (0.653) | −0.634 (1.261) | 0.100 (2.625) | $1.20 \times 10^{-1}$ |
| 0.254 | 3.198 (3.046) | 0.425 (0.937) | 1.164* (0.653) | −0.634 (1.261) | 0.100 (2.625) | $1.20 \times 10^{-1}$ |
| **1.064** | **3.198 (3.046)** | **0.425 (0.937)** | **1.164* (0.653)** | **−0.634 (1.261)** | **0.100 (2.625)** | $1.20 \times 10^{-1}$ |
| 0.228 | 3.198 (3.046) | 0.425 (0.937) | 1.164* (0.653) | −0.634 (1.261) | 0.100 (2.625) | $1.20 \times 10^{-1}$ |
| 1.168 | 3.198 (3.046) | 0.425 (0.937) | 1.164* (0.653) | −0.634 (1.261) | 0.100 (2.625) | $1.20 \times 10^{-1}$ |
| 0.768 | 3.198 (3.046) | 0.425 (0.937) | 1.164* (0.653) | −0.634 (1.261) | 0.100 (2.625) | $1.20 \times 10^{-1}$ |
| 1.378 | 3.198 (3.046) | 0.425 (0.937) | 1.164* (0.653) | −0.634 (1.261) | 0.100 (2.625) | $1.20 \times 10^{-1}$ |
| 1.088 | 3.198 (3.046) | 0.425 (0.937) | 1.164* (0.653) | −0.634 (1.261) | 0.100 (2.625) | $1.20 \times 10^{-1}$ |
| 0.770 | 3.198 (3.046) | 0.425 (0.937) | 1.164* (0.653) | −0.634 (1.261) | 0.100 (2.625) | $1.20 \times 10^{-1}$ |
| 0.607 | 3.198 (3.046) | 0.425 (0.937) | 1.164* (0.653) | −0.634 (1.261) | 0.100 (2.625) | $1.20 \times 10^{-1}$ |
| 0.444 | 3.198 (3.046) | 0.425 (0.937) | 1.164* (0.653) | −0.634 (1.261) | 0.100 (2.625) | $1.20 \times 10^{-1}$ |
| True | 5.000 | 1.000 | 1.000 | −1.000 | 1.000 | |

Source: author's calculations from the stored full estimates `q8e_blp_only_T10_J3` (per-start values recomputed at each stored $\sigma$ minimizer, no re-optimization). Bold: start with the lowest GMM objective across starts. (n.a.): estimated variance not positive; very large standard errors indicate weak identification..

Table 12: Convergence criterion and contraction error by starting point, T=10, J=3.

| Start $\sigma_0$ | $\hat{\sigma}_\alpha$ | GMM obj. | Converged | Iterations | $\|\nabla\|_\infty$ | Contraction error |
|---|---|---|---|---|---|---|
| 0.594 | 0.1000 | $1.20 \times 10^{-1}$ | yes | 3 | $2.82 \times 10^{-9}$ | $1.71 \times 10^{-6}$ |
| 0.908 | 0.1000 | $1.20 \times 10^{-1}$ | yes | 3 | $8.22 \times 10^{-9}$ | $1.71 \times 10^{-6}$ |
| 0.859 | 0.1000 | $1.20 \times 10^{-1}$ | yes | 3 | $7.01 \times 10^{-9}$ | $1.71 \times 10^{-6}$ |
| 1.426 | 0.1000 | $1.20 \times 10^{-1}$ | yes | 4 | $3.81 \times 10^{-11}$ | $1.71 \times 10^{-6}$ |
| 0.182 | 0.1000 | $1.20 \times 10^{-1}$ | yes | 3 | $1.48 \times 10^{-10}$ | $1.71 \times 10^{-6}$ |
| 0.876 | 0.1000 | $1.20 \times 10^{-1}$ | yes | 3 | $7.43 \times 10^{-9}$ | $1.71 \times 10^{-6}$ |
| 0.254 | 0.1000 | $1.20 \times 10^{-1}$ | yes | 3 | $4.00 \times 10^{-10}$ | $1.71 \times 10^{-6}$ |
| **1.064** | **0.1000** | $1.20 \times 10^{-1}$ | **yes** | **4** | $1.32 \times 10^{-11}$ | $1.71 \times 10^{-6}$ |
| 0.228 | 0.1000 | $1.20 \times 10^{-1}$ | yes | 3 | $2.95 \times 10^{-10}$ | $1.71 \times 10^{-6}$ |
| 1.168 | 0.1000 | $1.20 \times 10^{-1}$ | yes | 4 | $1.79 \times 10^{-11}$ | $1.71 \times 10^{-6}$ |
| 0.768 | 0.1000 | $1.20 \times 10^{-1}$ | yes | 3 | $5.17 \times 10^{-9}$ | $1.71 \times 10^{-6}$ |
| 1.378 | 0.1000 | $1.20 \times 10^{-1}$ | yes | 4 | $3.31 \times 10^{-11}$ | $1.71 \times 10^{-6}$ |
| 1.088 | 0.1000 | $1.20 \times 10^{-1}$ | yes | 4 | $1.42 \times 10^{-11}$ | $1.71 \times 10^{-6}$ |
| 0.770 | 0.1000 | $1.20 \times 10^{-1}$ | yes | 3 | $5.20 \times 10^{-9}$ | $1.71 \times 10^{-6}$ |
| 0.607 | 0.1000 | $1.20 \times 10^{-1}$ | yes | 3 | $2.96 \times 10^{-9}$ | $1.71 \times 10^{-6}$ |
| 0.444 | 0.1000 | $1.20 \times 10^{-1}$ | yes | 3 | $1.55 \times 10^{-9}$ | $1.71 \times 10^{-6}$ |

Source: author's calculations from the stored full estimates `q8e_blp_only_T10_J3` (per-start values recomputed at each stored $\sigma$ minimizer, no re-optimization); failed starts from `q8e_blp_only_T10_J3_failed_starts`. Bold: start with the lowest GMM objective across starts..

Table 13: Estimates with and without the Hausman instrument, T=10, J=3.

| | True | BLP | BLP + Hausman |
|---|---|---|---|
| Constant | 5.000 | 3.198 | 4.101*** |
| | | (3.046) | (1.235) |
| $x_2$ (uniform) | 1.000 | 0.425 | 0.495 |
| | | (0.937) | (0.903) |
| $x_3$ (normal) | 1.000 | 1.164* | 1.012*** |
| | | (0.653) | (0.331) |
| Price coefficient | −1.000 | −0.634 | −0.993*** |
| | | (1.261) | (0.330) |
| $\sigma_\alpha$ | 1.000 | 0.100 | 0.100 |
| | | (2.625) | (1.945) |
| GMM objective | | $1.20 \times 10^{-1}$ | $6.06 \times 10^{-1}$ |
| Observations ($JT$) | | 30 | 30 |

Source: author's calculations from the stored full estimates `q8e_blp_only_T10_J3` and `q8e_blp_hausman_T10_J3`; best start of each. (n.a.): estimated variance not positive; very large standard errors indicate weak identification..

## 8f

Table 14: Estimates imposing $E[\xi_{jt}p_{jt}] = 0$ (price as its own instrument) versus the original IV specification.

| | True | BLP (IV) | Price as instrument |
|---|---|---|---|
| Constant | 5.000 | 5.589 | $4.775^{***}$ |
| | | $(1.62 \times 10^7)$ | (0.352) |
| $x_2$ (uniform) | 1.000 | 1.100 | $0.922^{***}$ |
| | | $(3.58 \times 10^6)$ | (0.216) |
| $x_3$ (normal) | 1.000 | 1.082 | $1.022^{***}$ |
| | | $(1.20 \times 10^6)$ | (0.078) |
| Price coefficient | −1.000 | −1.168 | $-0.904^{***}$ |
| | | $(5.27 \times 10^6)$ | (0.083) |
| $\sigma_\alpha$ | 1.000 | 1.054 | $0.989^{***}$ |
| | | $(1.24 \times 10^6)$ | (0.085) |
| GMM objective | | $5.23 \times 10^{-3}$ | $5.83 \times 10^{-3}$ |
| Observations ($JT$) | | 300 | 300 |

Source: author's calculations from the stored full estimates `q8c_blp_only_T100_J3` (original) and `q8f_blp_price_T100_J3`.

Table 15: Price-as-instrument estimates with and without the Hausman instrument.

| | True | Price as instrument | Price + Hausman |
|---|---|---|---|
| Constant | 5.000 | $4.775^{***}$ | $4.867^{***}$ |
| | | (0.352) | (0.341) |
| $x_2$ (uniform) | 1.000 | $0.922^{***}$ | $0.945^{***}$ |
| | | (0.216) | (0.220) |
| $x_3$ (normal) | 1.000 | $1.022^{***}$ | $1.036^{***}$ |
| | | (0.078) | (0.076) |
| Price coefficient | −1.000 | $-0.904^{***}$ | $-0.925^{***}$ |
| | | (0.083) | (0.082) |
| $\sigma_\alpha$ | 1.000 | $0.989^{***}$ | $1.015^{***}$ |
| | | (0.085) | (0.080) |
| GMM objective | | $5.83 \times 10^{-3}$ | $3.69 \times 10^{-1}$ |
| Observations ($JT$) | | 300 | 300 |

Source: author's calculations from the stored full estimates `q8f_blp_price_T100_J3` and `q8f_blp_price_hausman_T100_J3`.

# Question 9: Cost shifters and the supply side

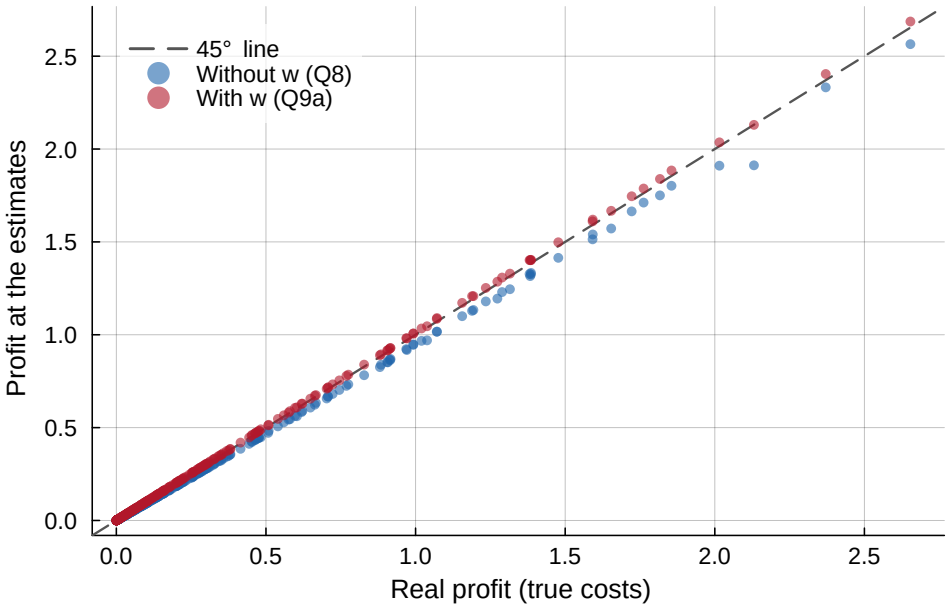
## 9a

Table 16: Demand estimates with and without the cost shifter $w_j$ in the instrument set (standard errors in parentheses; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$).

| | True | BLP (Q8) | BLP $+w$ (Q9a) | BLP + Haus. (Q8) | BLP + Haus. $+w$ (Q9a) |
|---|---|---|---|---|---|
| Constant | 5.000 | 5.589 | 5.012*** | 4.983*** | 5.005*** |
| | | $(1.62 \times 10^7)$ | (0.396) | (0.404) | (0.397) |
| $x_2$ (uniform) | 1.000 | 1.100 | 0.973*** | 0.967*** | 0.971*** |
| | | $(3.58 \times 10^6)$ | (0.227) | (0.225) | (0.226) |
| $x_3$ (normal) | 1.000 | 1.082 | 1.039*** | 1.037*** | 1.037*** |
| | | $(1.20 \times 10^6)$ | (0.076) | (0.076) | (0.076) |
| Price coefficient | −1.000 | −1.168 | −0.981*** | −0.972*** | −0.980*** |
| | | $(5.27 \times 10^6)$ | (0.113) | (0.116) | (0.113) |
| $\sigma_\alpha$ | 1.000 | 1.054 | 1.009*** | 1.007*** | 1.005*** |
| | | $(1.24 \times 10^6)$ | (0.080) | (0.080) | (0.080) |
| GMM objective | | $5.23 \times 10^{-3}$ | $5.52 \times 10^{-3}$ | $5.56 \times 10^{-3}$ | $2.00 \times 10^{-1}$ |
| Observations ($JT$) | | 300 | 300 | 300 | 300 |

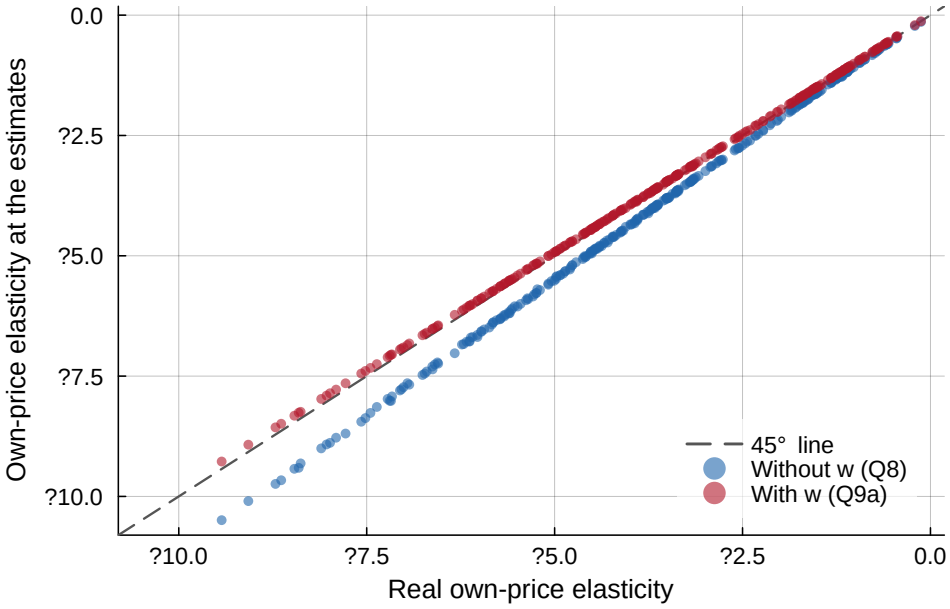
Source: author's calculations from the stored full estimates `q8c_*` and `q9a_*` (T=100, J=3).

Figure 11: Product-level profits at the estimates, with and without the cost shifter, versus real profits.




Source: author's calculations from the stored full estimates `q8c_*` and `q9a_*` (T=100, J=3); profits $(p - \hat{c})s$ with costs recovered under Bertrand pricing.

Figure 12: Own-price elasticities at the estimates, with and without the cost shifter, versus real elasticities.




Source: author's calculations from the stored full estimates `q8c_*` and `q9a_*` (T=100, J=3).

## 9b

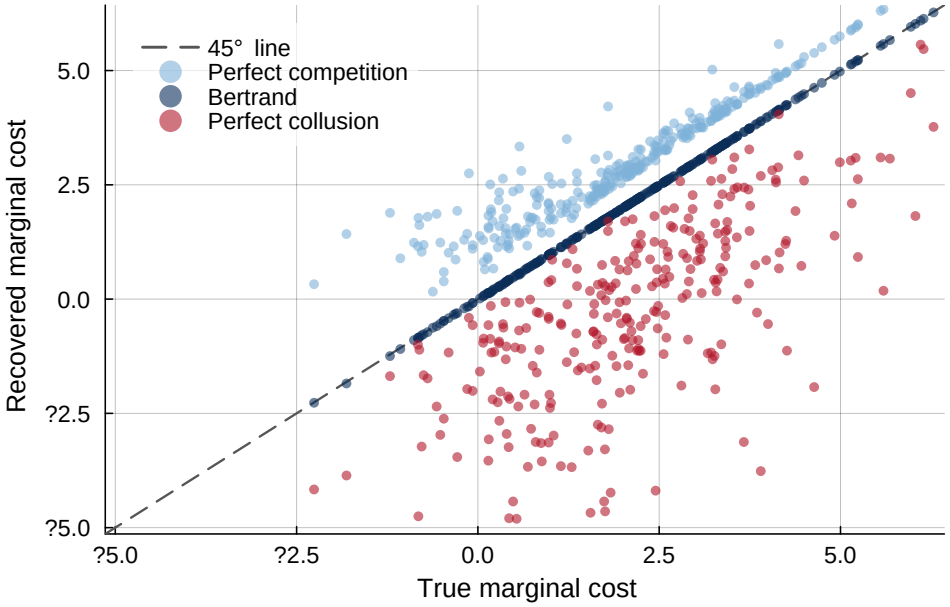
Table 17: Small-sample estimates (T=10, J=3) with and without the cost shifter $w_j$ (standard errors in parentheses; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$).

| | True | BLP (Q8e) | BLP $+w$ (Q9b) | BLP + Haus. (Q8e) | BLP + Haus. $+w$ (Q9b) |
|---|---|---|---|---|---|
| Constant | 5.000 | 3.198 | 1.878 | 4.101*** | 3.103** |
| | | (3.046) | (3.068) | (1.235) | (1.545) |
| $x_2$ (uniform) | 1.000 | 0.425 | 0.350 | 0.495 | 0.417 |
| | | (0.937) | (1.352) | (0.903) | (0.985) |
| $x_3$ (normal) | 1.000 | 1.164* | 1.498** | 1.012*** | 1.184*** |
| | | (0.653) | (0.626) | (0.331) | (0.357) |
| Price coefficient | −1.000 | −0.634 | 0.069 | −0.993*** | −0.591 |
| | | (1.261) | (1.139) | (0.330) | (0.387) |
| $\sigma_\alpha$ | 1.000 | 0.100 | 0.605 | 0.100 | 0.138 |
| | | (2.625) | (1.484) | (1.945) | (1.436) |
| GMM objective | | $1.20 \times 10^{-1}$ | $5.51 \times 10^{0}$ | $6.06 \times 10^{-1}$ | $6.73 \times 10^{0}$ |
| Observations ($JT$) | | 30 | 30 | 30 | 30 |

Source: author's calculations from the stored full estimates `q8e_*` and `q9b_*` (T=10, J=3).

**9d**

Figure 13: True versus recovered marginal costs under perfect competition, Bertrand pricing and perfect collusion (axes cover the 1st to 99th percentile of the plotted values).




Source: author's calculations: cost inversion at the demand estimates of `q9a_blp_hausman_w_T100_J3`.

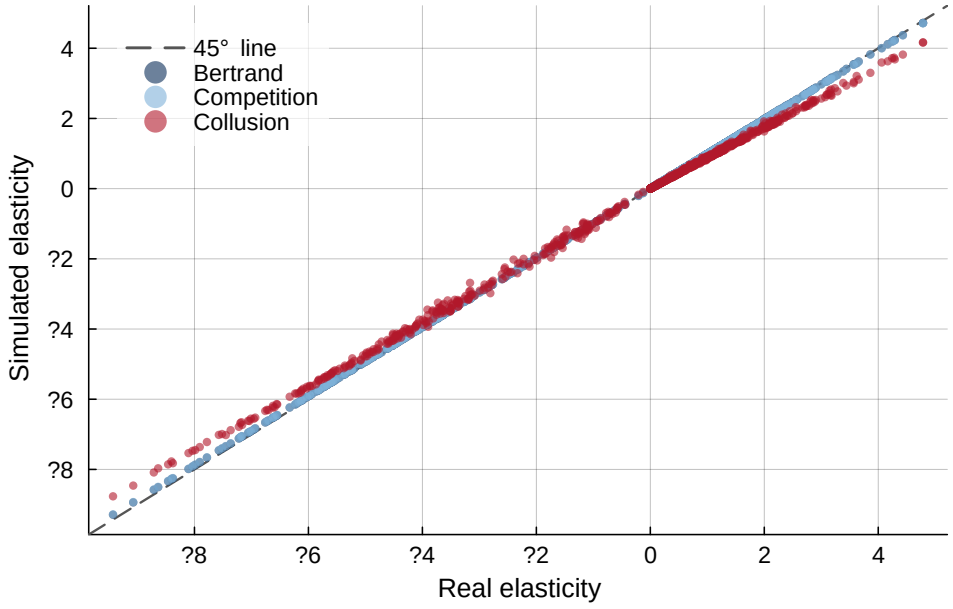
**9e**

Table 18: Joint demand and supply estimates under three conduct assumptions (standard errors in parentheses; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$).

| | True | Bertrand | Competition | Collusion |
|---|---|---|---|---|
| Constant | 5.000 | 5.019*** | 5.005*** | 4.109*** |
| | | (0.397) | (0.397) | (0.966) |
| $x_2$ (uniform) | 1.000 | 0.975*** | 0.971*** | 0.778*** |
| | | (0.226) | (0.226) | (0.261) |
| $x_3$ (normal) | 1.000 | 1.039*** | 1.037*** | 0.895*** |
| | | (0.076) | (0.076) | (0.137) |
| Price coefficient | −1.000 | −0.983*** | −0.980*** | −0.781*** |
| | | (0.113) | (0.113) | (0.251) |
| $\gamma_0$ (cost constant) | 2.000 | 1.940*** | 2.951*** | −1.152*** |
| | | (0.074) | (0.055) | (0.437) |
| $\gamma_z$ ($z_{jt}$) | 1.000 | 0.916*** | 0.790*** | 0.598 |
| | | (0.056) | (0.051) | (0.555) |
| $\gamma_w$ ($w_j$) | 1.000 | 0.979*** | 0.776*** | 0.622 |
| | | (0.075) | (0.069) | (0.677) |
| $\sigma_\alpha$ | 1.000 | 1.009*** | 1.005*** | 0.701** |
| | | (0.079) | (0.080) | (0.312) |
| GMM objective | | $3.15 \times 10^0$ | $8.09 \times 10^0$ | $2.86 \times 10^1$ |
| Observations ($JT$) | | 300 | 300 | 300 |

Source: author's calculations from the stored full estimates `q9e_system_*_T100_J3`.

Figure 14: Real versus simulated price elasticities (all own- and cross-price elements) under the three conduct assumptions.



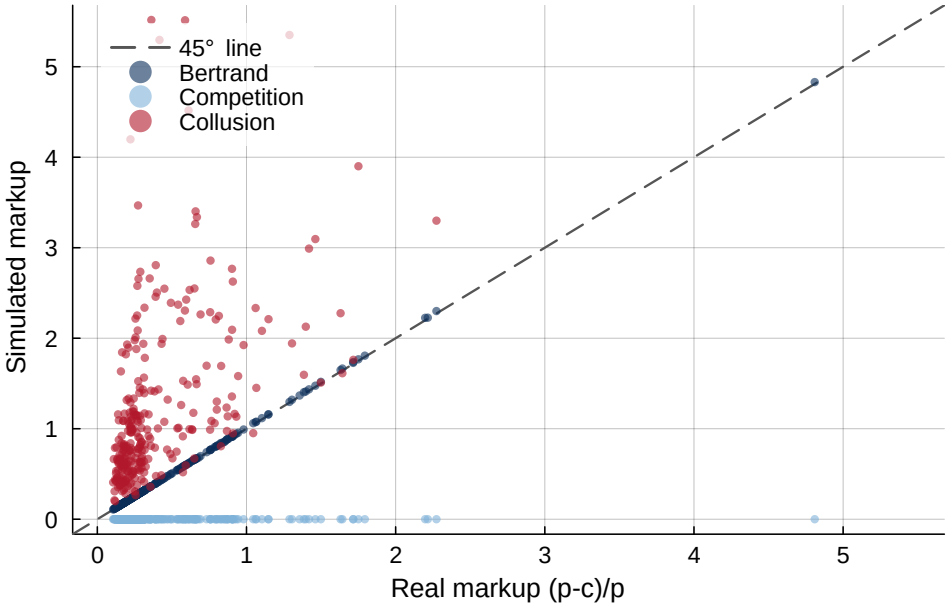

Source: author's calculations from the stored full estimates `q9e_system_*_T100_J3`.

Table 19: Descriptive statistics of real and simulated elasticities.

| | Mean | SD | Median | Q25 | Q75 |
|---|---|---|---|---|---|
| *Own-price elasticities* | | | | | |
| Real | −3.709 | 2.080 | −3.650 | −5.084 | −1.744 |
| Bertrand | −3.661 | 2.049 | −3.613 | −5.012 | −1.722 |
| Competition | −3.658 | 2.047 | −3.609 | −5.008 | −1.722 |
| Collusion | −3.516 | 1.930 | −3.448 | −4.844 | −1.705 |
| *Cross-price elasticities* | | | | | |
| Real | 1.003 | 0.947 | 0.771 | 0.210 | 1.475 |
| Bertrand | 0.992 | 0.937 | 0.761 | 0.207 | 1.459 |
| Competition | 0.990 | 0.935 | 0.760 | 0.207 | 1.456 |
| Collusion | 0.876 | 0.822 | 0.676 | 0.184 | 1.303 |

Source: author's calculations from the stored full estimates `q9e_system_*_T100_J3`.

Figure 15: Real versus simulated markups under the three conduct assumptions (axes cover the 1st to 99th percentile of the plotted values).




Source: author's calculations from the stored full estimates `q9e_system_*_T100_J3`.

Table 20: Descriptive statistics of real and simulated markups.

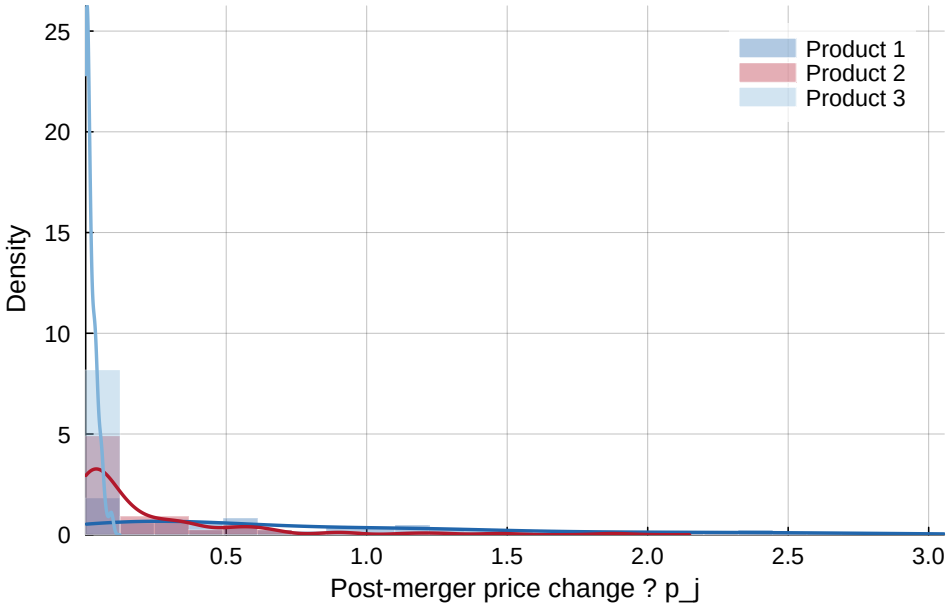
| | Mean | SD | Median | Q25 | Q75 |
|---|---|---|---|---|---|
| Real | 0.472 | 0.634 | 0.274 | 0.197 | 0.574 |
| Bertrand | 0.477 | 0.636 | 0.277 | 0.200 | 0.581 |
| Competition | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 |
| Collusion | 4.275 | 35.432 | 0.988 | 0.629 | 1.694 |

Source: author's calculations from the stored full estimates `q9e_system_*_T100_J3`.

## Question 10: Merger simulation

### 10b

Figure 16: Distribution and kernel density of the post-merger price changes of products 1, 2 (merging) and 3 (non-merging) across markets.




Source: author's calculations: merger counterfactual (firms 1 and 2, no cost efficiencies) at the estimates of `q9e_system_bertrand_T100_J3`.

Table 21: Descriptive statistics of post-merger price changes by product.

| Product | Mean | SD | Median | Q25 | Q75 |
|---|---|---|---|---|---|
| 1 | 0.800 | 0.771 | 0.539 | 0.170 | 1.204 |
| 2 | 0.213 | 0.329 | 0.075 | 0.017 | 0.270 |
| 3 | 0.019 | 0.021 | 0.012 | 0.002 | 0.032 |

Source: author's calculations: merger counterfactual (firms 1 and 2, no cost efficiencies) at the estimates of `q9e_system_bertrand_T100_J3`.

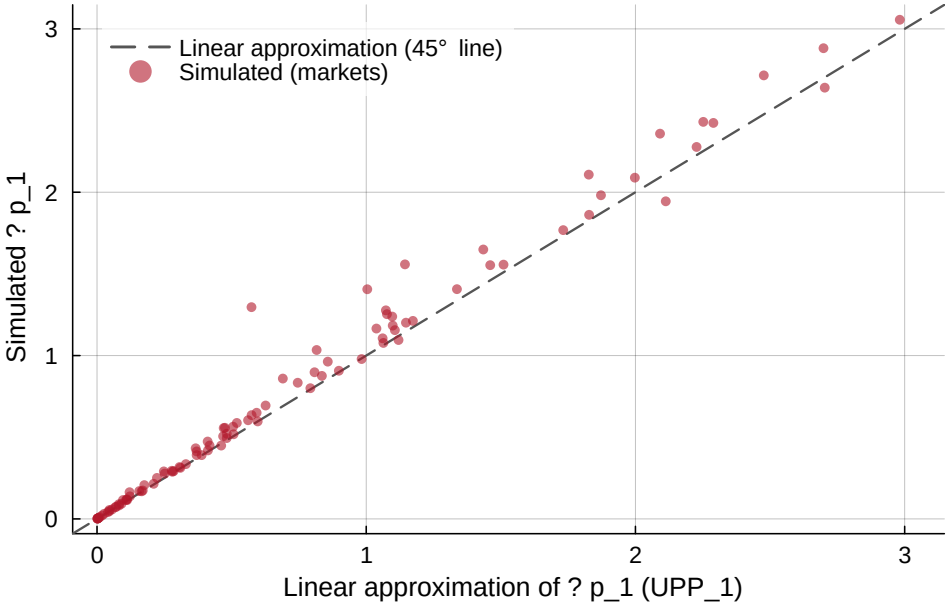
Table 22: Descriptive statistics of the post-merger changes in consumer, producer and total surplus across markets.

| | Mean | SD | Median | Q25 | Q75 |
|---|---|---|---|---|---|
| $\Delta$ Consumer surplus | −0.084 | 0.147 | −0.029 | −0.092 | −0.008 |
| $\Delta$ Producer surplus | 0.045 | 0.078 | 0.017 | 0.004 | 0.048 |
| $\Delta$ Total surplus | −0.039 | 0.074 | −0.009 | −0.043 | −0.002 |

Source: author's calculations: merger counterfactual (firms 1 and 2, no cost efficiencies) at the estimates of `q9e_system_bertrand_T100_J3`.

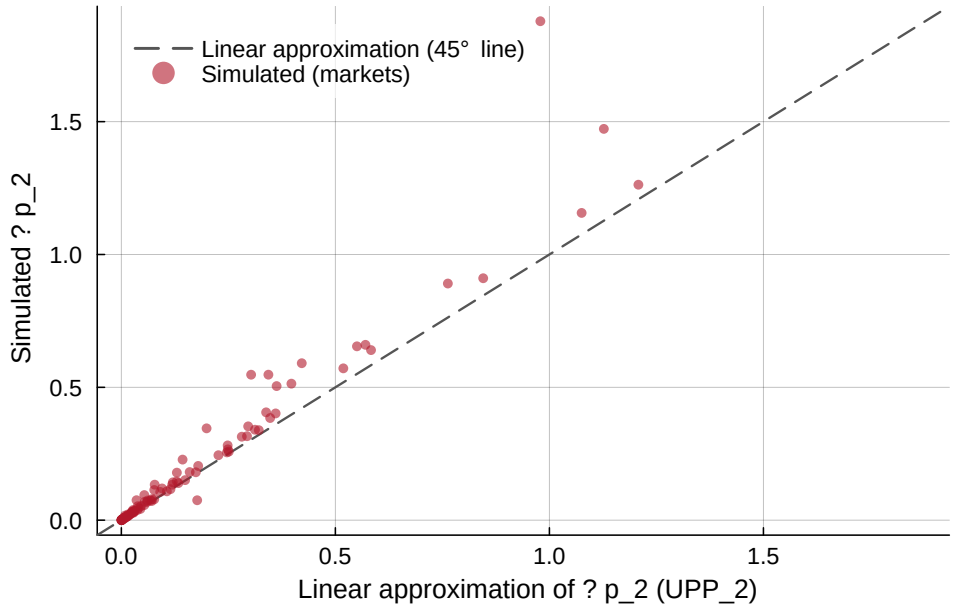
**10c**

Figure 17: Simulated price change of product 1 against upward pricing pressure, one point per market.



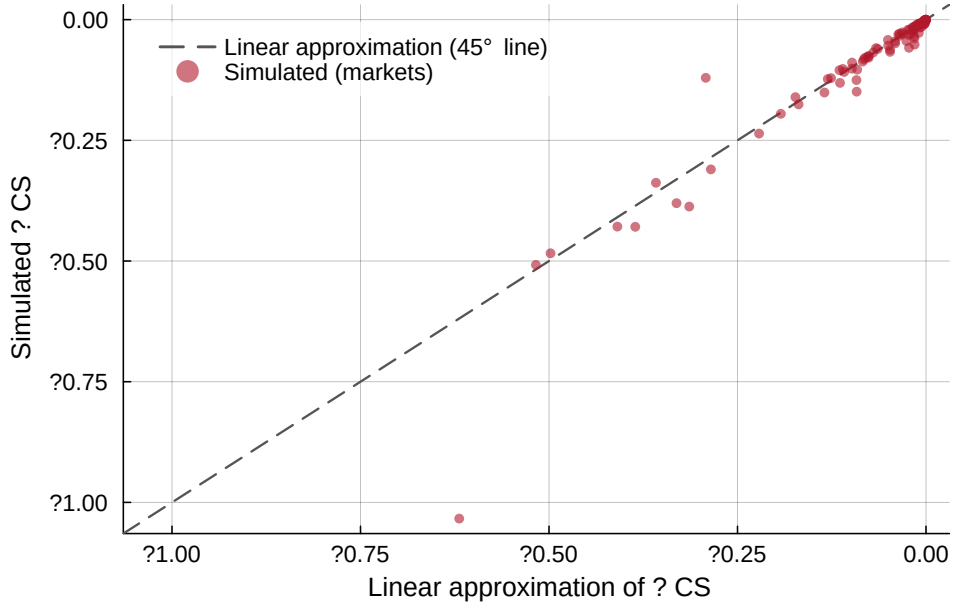

Source: author's calculations: merger counterfactual (firms 1 and 2, no cost efficiencies) at the estimates of `q9e_system_bertrand_T100_J3`; diversion ratios and $\text{UPP}_j = D_{jk}(p_k - c_k)$ from the pre-merger estimates.

Figure 18: Simulated price change of product 2 against upward pricing pressure, one point per market.



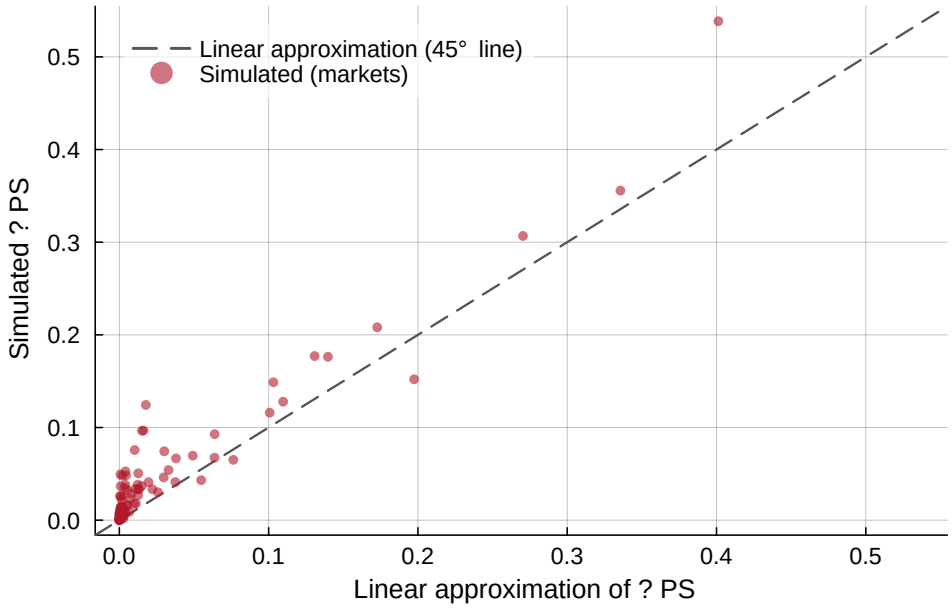

Source: author's calculations: merger counterfactual (firms 1 and 2, no cost efficiencies) at the estimates of `q9e_system_bertrand_T100_J3`; diversion ratios and $\text{UPP}_j = D_{jk}(p_k - c_k)$ from the pre-merger estimates.

Figure 19: Simulated change in consumer surplus against its linear approximation, one point per market.



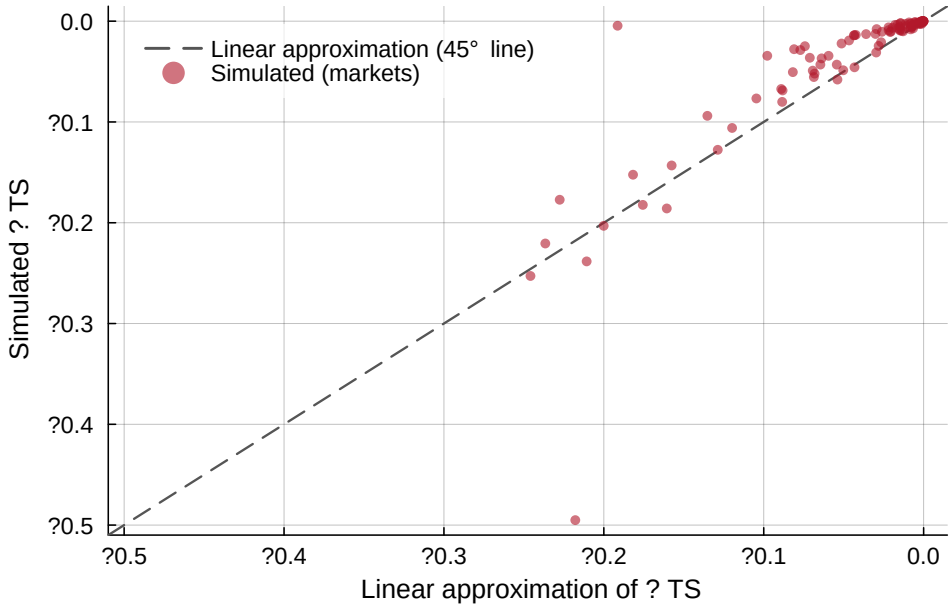

Source: author's calculations: merger counterfactual (firms 1 and 2, no cost efficiencies) at the estimates of `q9e_system_bertrand_T100_J3`; diversion ratios and $\text{UPP}_j = D_{jk}(p_k - c_k)$ from the pre-merger estimates.

Figure 20: Simulated change in producer surplus against its linear approximation, one point per market.




Source: author's calculations: merger counterfactual (firms 1 and 2, no cost efficiencies) at the estimates of `q9e_system_bertrand_T100_J3`; diversion ratios and $\text{UPP}_j = D_{jk}(p_k - c_k)$ from the pre-merger estimates.

Figure 21: Simulated change in total surplus against its linear approximation, one point per market.




Source: author's calculations: merger counterfactual (firms 1 and 2, no cost efficiencies) at the estimates of `q9e_system_bertrand_T100_J3`; diversion ratios and $\text{UPP}_j = D_{jk}(p_k - c_k)$ from the pre-merger estimates.

## Question 11: Newer instruments and substitution diagnostics

### 11a

Table 23: Price first stage with each instrument family, T=100, J=5 (heteroskedasticity-robust standard errors in parentheses; $F$ tests the excluded instruments given $x$).

| Dependent variable: price | Hausman | BLP | GH local | GH quadratic | All |
|---|---|---|---|---|---|
| Constant | 0.124 | 2.030*** | 2.258*** | 1.805*** | 0.093 |
| | (0.211) | (0.292) | (0.205) | (0.180) | (0.428) |
| $x_2$ | −0.091 | −0.217 | −0.168 | −0.175 | −0.068 |
| | (0.205) | (0.226) | (0.231) | (0.229) | (0.210) |
| $x_3$ | −0.009 | −0.061 | −0.069 | −0.067 | −0.019 |
| | (0.057) | (0.063) | (0.064) | (0.063) | (0.058) |
| Hausman $\bar{p}_{j,-t}$ | 0.959*** | | | | 0.958*** |
| | (0.088) | | | | (0.090) |
| BLP rival sum $x_2$ | | −0.012 | | | −0.040 |
| | | (0.127) | | | (0.114) |
| BLP rival sum $x_3$ | | 0.013 | | | −0.001 |
| | | (0.031) | | | (0.029) |
| GH local rival $x_2$ | | | −0.044 | | 0.077 |
| | | | (0.061) | | (0.077) |
| GH local rival $x_3$ | | | −0.089 | | −0.086 |
| | | | (0.058) | | (0.077) |
| GH quadratic rival $x_2$ | | | | 0.190 | 0.226 |
| | | | | (0.136) | (0.178) |
| GH quadratic rival $x_3$ | | | | 0.008 | −0.001 |
| | | | | (0.009) | (0.013) |
| $R^2$ | 0.200 | 0.004 | 0.009 | 0.009 | 0.207 |
| $F$ (excluded instruments) | 121.7 | 0.1 | 1.3 | 1.3 | 17.9 |
| Observations | 500 | 500 | 500 | 500 | 500 |

Source: author's calculations from `100markets5products.mat`; first-stage regressions include a constant, $x_2$ and $x_3$.

Table 24: Estimates and final GMM objective by instrument set, T=100, J=5 (standard errors in parentheses; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$).

| | True | BLP | GH | GH + Hausman |
|---|---|---|---|---|
| Constant | 5.000 | 3.877 | 5.538*** | 5.295*** |
| | | (n.a.) | (1.561) | (0.641) |
| $x_2$ (uniform) | 1.000 | 0.860 | 0.687* | 0.706*** |
| | | (n.a.) | (0.367) | (0.173) |
| $x_3$ (normal) | 1.000 | 1.039 | 0.851*** | 1.007*** |
| | | (n.a.) | (0.119) | (0.090) |
| Price coefficient | −1.000 | −0.422 | −1.562* | −1.100*** |
| | | (n.a.) | (0.812) | (0.076) |
| $\sigma_\alpha$ | 1.000 | 0.927 | 0.100 | 0.950*** |
| | | (n.a.) | (3.118) | (0.208) |
| GMM objective | | $6.28 \times 10^{-1}$ | $2.58 \times 10^{-1}$ | $3.56 \times 10^{0}$ |
| Observations ($JT$) | | 500 | 500 | 500 |

Source: author's calculations from the stored full estimates `q11a_*_T100_J5`; best start of each. No fixed effects are used in any specification. (n.a.): estimated variance not positive; very large standard errors indicate weak identification..

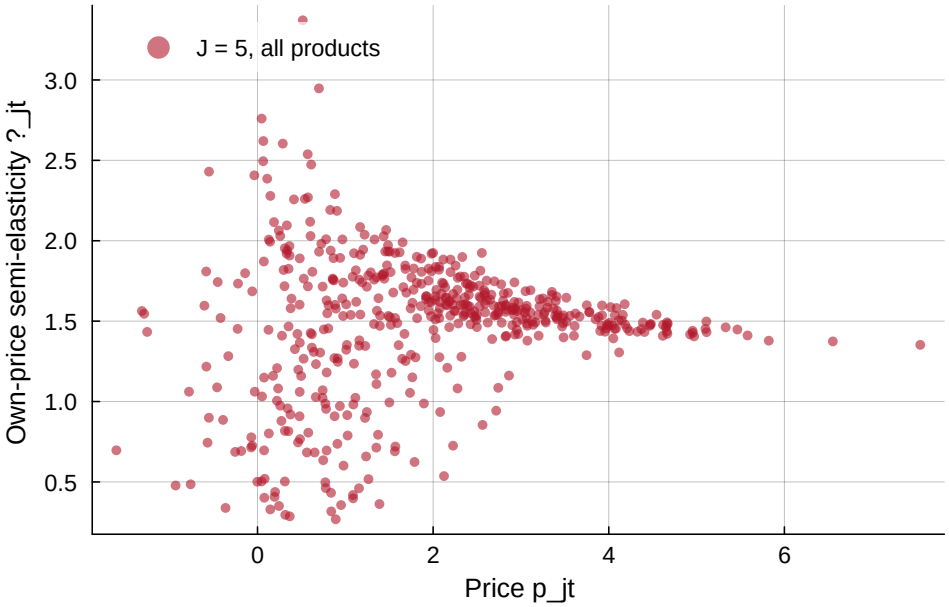
Table 25: Sensitivity to starting values, T=100, J=5.

| Instruments | Starts ok | Starts failed | Converged | Min $\hat{\sigma}_\alpha$ | Max $\hat{\sigma}_\alpha$ | Min obj. | Max obj. | Mean iter. |
|---|---|---|---|---|---|---|---|---|
| BLP | 7 | 9 | 7 | 0.9272 | 0.9272 | $6.28 \times 10^{-1}$ | $6.28 \times 10^{-1}$ | 3.3 |
| GH | 16 | 0 | 3 | 0.1000 | 0.3727 | $2.58 \times 10^{-1}$ | $4.38 \times 10^{-1}$ | 1.6 |
| GH + Hausman | 6 | 10 | 6 | 0.9504 | 0.9504 | $3.56 \times 10^{0}$ | $3.56 \times 10^{0}$ | 3.7 |

Source: author's calculations from the stored full estimates `q11a_*_T100_J5` and the corresponding `*_failed_starts` files; summary over all starting points.

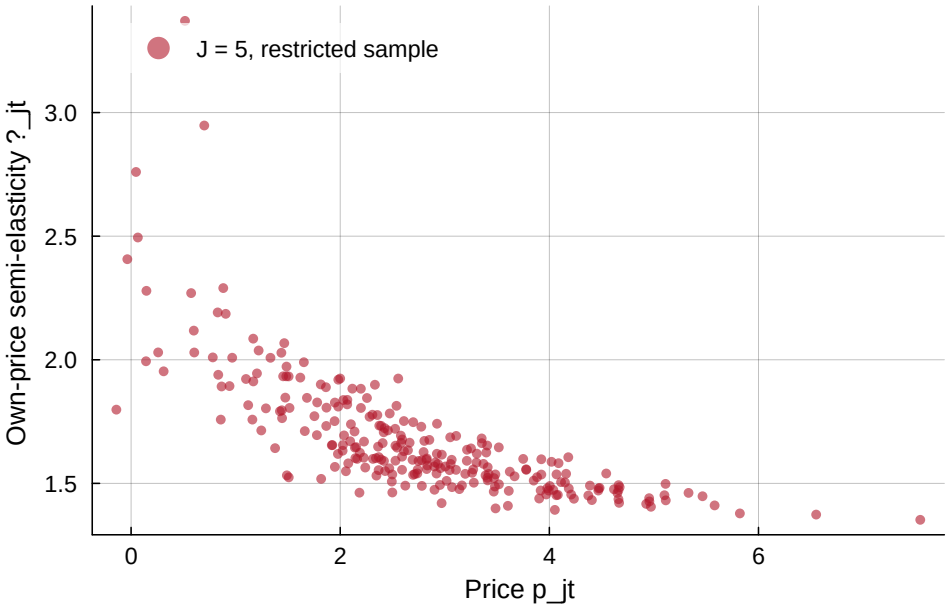
**11c**

Figure 22: Own-price semi-elasticity against price, all product-markets.




Source: author's calculations at the stored estimates `q11a_gh_hausman_T100_J5`; $\eta_{jt} = -s_{jt}^{-1}\partial s_{jt}/\partial p_{jt}$.

Figure 23: Own-price semi-elasticity against price, restricted sample with small approximation error ($|i_{jt}^{scp}/\eta_{jt}| < 0.25$).




Source: author's calculations at the stored estimates `q11a_gh_hausman_T100_J5`; $\eta_{jt} = -s_{jt}^{-1}\partial s_{jt}/\partial p_{jt}$; approximation-error ratio as in `pset_io`.

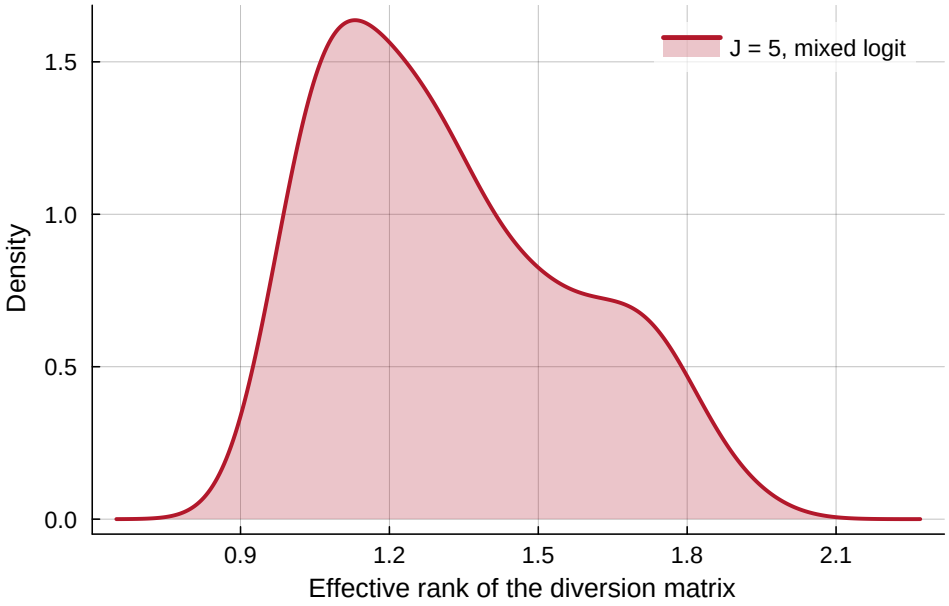
Table 26: Flexible regression of the own-price semi-elasticity on price and characteristics (the same specification in every column). Columns differ only in the sample: the cut-off on $|i_{jt}^{scp}/\eta_{jt}|$ falls from left to right, so the sample becomes more restricted (heteroskedasticity-robust standard errors in parentheses; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$).

| Cut-off on $\lvert i_{jt}^{scp}/\eta_{jt}\rvert$ | Full | 2.00 | 1.00 | 0.75 | 0.50 | 0.25 | 0.10 |
|---|---|---|---|---|---|---|---|
| Constant | $1.603^{***}$ | $1.885^{***}$ | $2.037^{***}$ | $2.081^{***}$ | $2.236^{***}$ | $2.381^{***}$ | $2.294^{***}$ |
| | (0.117) | (0.106) | (0.096) | (0.093) | (0.103) | (0.097) | (0.129) |
| $p$ | $0.103^{**}$ | $-0.078^{*}$ | $-0.163^{***}$ | $-0.192^{***}$ | $-0.274^{***}$ | $-0.339^{***}$ | $-0.295^{***}$ |
| | (0.046) | (0.045) | (0.042) | (0.042) | (0.048) | (0.044) | (0.061) |
| $x_2$ | $-0.858^{***}$ | $-0.417^{*}$ | $-0.291$ | $-0.272$ | $-0.341^{*}$ | $-0.358^{*}$ | $-0.363$ |
| | (0.295) | (0.225) | (0.190) | (0.185) | (0.187) | (0.209) | (0.238) |
| $x_3$ | $-0.227^{***}$ | $-0.191^{***}$ | $-0.132^{***}$ | $-0.131^{***}$ | $-0.131^{***}$ | $-0.088^{*}$ | $-0.086$ |
| | (0.039) | (0.034) | (0.038) | (0.040) | (0.044) | (0.047) | (0.055) |
| $p^2$ | $-0.025^{***}$ | $-0.002$ | $0.007$ | $0.012^{**}$ | $0.023^{***}$ | $0.028^{***}$ | $0.024^{***}$ |
| | (0.006) | (0.005) | (0.006) | (0.006) | (0.006) | (0.006) | (0.007) |
| $1/p$ | $0.002^{***}$ | $0.009$ | $0.014$ | $0.013$ | $0.012$ | $0.005$ | $0.020^{**}$ |
| | (0.000) | (0.014) | (0.014) | (0.012) | (0.010) | (0.007) | (0.009) |
| $x_2^2$ | $0.561^{**}$ | $0.248$ | $0.093$ | $0.114$ | $0.193$ | $0.100$ | $0.143$ |
| | (0.231) | (0.154) | (0.125) | (0.121) | (0.119) | (0.134) | (0.155) |
| $x_3^2$ | $-0.007$ | $-0.010$ | $-0.011$ | $-0.003$ | $-0.008$ | $-0.011$ | $-0.007$ |
| | (0.013) | (0.011) | (0.011) | (0.014) | (0.013) | (0.013) | (0.013) |
| $p \cdot x_2$ | $0.081^{*}$ | $0.044$ | $0.056$ | $0.041$ | $0.040$ | $0.074^{**}$ | $0.056^{*}$ |
| | (0.047) | (0.041) | (0.039) | (0.039) | (0.033) | (0.033) | (0.033) |
| $p \cdot x_3$ | $0.052^{***}$ | $0.044^{***}$ | $0.029^{**}$ | $0.033^{**}$ | $0.032^{**}$ | $0.019$ | $0.019$ |
| | (0.013) | (0.011) | (0.012) | (0.013) | (0.014) | (0.015) | (0.017) |
| $R^2$ | 0.186 | 0.312 | 0.481 | 0.542 | 0.651 | 0.689 | 0.662 |
| Observations | 500 | 414 | 368 | 347 | 312 | 254 | 194 |

Source: author's calculations at the stored estimates `q11a_gh_hausman_T100_J5`; $\eta_{jt} = -s_{jt}^{-1}\partial s_{jt}/\partial p_{jt}$; each column keeps the product-markets with $|i_{jt}^{scp}/\eta_{jt}|$ below the cut-off in the header.

## 11d

Figure 24: Density of the effective rank (exponential of the entropy of normalized singular values) of the model-implied diversion matrix across markets.




Source: author's calculations at the stored estimates `q11a_gh_hausman_T100_J5`; mixed logit only.